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NGE 311 — Past Questions & Materials
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Course: NGE 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — NGE 311 OIL AND GAS SEPARATION PROCESSES

Level 300 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF NATURAL GAS ENGINEERING**

2024/2025 First Semester Examination

NGE 311: Oil and Gas Separation Processes.

Answer Question 1 and Any Other Three Questions. Time Allowed: 2HR:30MINS

Please write your matriculation Number boldly on your answer Booklet. Date: 26/05/2025

QUESTION 1 (19 Marks)

✕(1a) State three classification of Oilfield Separators and their possible dimension(diameter).

✓(1b) (i) Size a standard oil-gas two phase horizontal separator with the following conditions:

Gas capacity = 10 MMSCFD, Liquid capacity = 2000 bpd, Droplet size = 120 micron, Density of liquid = 50 lb/ft³, Density of gas = 3 lb/ft³, Drag coefficient = 0.75, Operating pressure = 1200 psia, Operating Temperature = 100 °F and Compressibility factor = 0.92.

✕Calculate the terminal settling velocity

✕(1c) Mention seven parameters used for separator design.

QUESTION 2 (17 Marks)

✓(2a) What are the main reasons for flashing natural gas?

(2b) The vapor compositions of a feed are 72%, 24% and 4% with an operating temperature of 298 K and pressure of 250 bar. If the feed has 0.85 amount of vapor with and with the following liquid/vapor equilibrium ratio of 60.8, 35.6 and 0.0396.

Determine the following:

✕(i) Liquid compositions (ii) feed compositions (iii) the amount of liquid in the feed.

✓(2c) Explain Separation techniques.

QUESTION 3 (17 Marks)

✕(3a) (i) In a tabular form, State three primary functions of an oil/gas separator, along with separation methods, internal devices and separation aid.

✓(ii) State and explain three reasons why we must separate water from natural gas.

(3b) Explain the following terms:

✕(i) Retention time (ii) Terminal settling velocity (iii) Gas capacity

✓(3c) State and explain the three main factors affecting oil and gas Separation.

QUESTION 4 (17 Marks) ✕

- 4a Define the following terms: (i) seam to seam length (ii) Liquid capacity (iii) Effective length
 (4b) What is the main function of slenderness ratio in sizing a separator?
 (4c) (i) Write brief note on vertical and horizontal separators.
 (ii) Draw and label a typical three-stage compressor train separators in the Oil and Gas Industry.
 (iii) Differentiate between Monotube vs. Dual Tube Oilfield Separators

QUESTION 5 (17 Marks)

- (5a) Draw and label a two phase horizontal separator
 (5b) State two advantages and disadvantages of vertical and horizontal separator
 (5c)(i) How can you separate natural gas from water?
 (ii) The feed to a high pressure separator enters at 10,000 kg/hr and has the following compositions: C₁80%, C₂ = 10%, C₃ = 7%, C₄=3%. the overhead is produced at 2800kg/hr and contains 90% of C₁, 5% of C₂, 3% of C₃ and the rest C₄ +. Calculate the weight of the bottom stream and each composition.

Hints: $F = L + V$, $Fz_i = Lx_i + Vy_i$.

$$L_{ss} = + \frac{4}{3} L_{eff} \quad dL_{eff} = 420 \left(\frac{TZQ_g}{P} \right) \left[\left(\frac{\rho_g}{\rho_l - \rho_g} \right) \frac{C_D}{d_m} \right]^{1/2} \quad d^2 L_{eff} = \frac{t_{tr} Q_L}{0.7}$$

$$y_i = \frac{x_i p^{sat}(T)}{P} \quad L_{ss} = L_{eff} + \frac{d}{12} \quad S_R = \frac{12L_{ss}}{d} \quad k_i = \frac{(P_i)^{sat}(T)}{P} = \frac{y_i}{x_i}$$

$$p^{sat}(T) = e^{[A - \frac{B}{T+C}]} \quad \sum_i \frac{y_i}{(P_i)^{sat}(T)} = \frac{1}{P} \quad x_i = \frac{y_i P}{p^{sat}(T)}$$

$$x_i = \frac{z_i}{1 + \frac{V}{F}(k_i - 1)} \quad V_t = 0.0199 \left[\left(\frac{\rho_l - \rho_g}{\rho_g} \right) \frac{d_m}{C_D} \right]^{1/2} \quad \sum_i x_i (P_i)^{sat}(T) = P$$